

Part 1A — Intro to Analysis

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These notes are not endorsed by the lecturers, and I have modified them (often significantly) after lectures. They are nowhere near accurate representations of what was actually lectured, and in particular, all errors are almost surely mine.

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0 Introduction

Calculus has two branches. *Differential calculus* studies slopes and tangent lines; *integral calculus* studies areas and accumulation. Both rest on the notion of a limit, and limits only behave well because of a property of the real numbers that the rationals lack. This course therefore starts by pinning down exactly what the real numbers are.

We first recall the number systems $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ and the language of statements and quantifiers in which everything else is written, together with induction and the well-ordering principle, the two basic tools for proving statements about all natural numbers.

We then describe $\mathbb{R}$ as a *complete ordered field*. The field axioms let us compute, the order axioms let us compare, and the completeness axiom — every nonempty set that is bounded above has a least upper bound — is what separates $\mathbb{R}$ from $\mathbb{Q}$. Most of the first chapter is about what completeness buys us: the Archimedean property, the existence of $\sqrt{2}$, and the density of $\mathbb{Q}$ and of the irrationals in $\mathbb{R}$.

Finally, we define what it means for a sequence of real numbers to converge. This is the first genuine limit in the course, and the ϵ -arguments that appear there are the pattern for everything that follows.

1 The real number system

In this chapter we set up the language of quantifiers and induction, and then describe $\mathbb{R}$ axiomatically as a complete ordered field. The completeness axiom is the one that matters for analysis, and most of the chapter is about its consequences.

1.1 Numbers and logic

The number systems we start from are

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

$\mathbb{N}$ is closed under addition, and $\mathbb{Z}$ under addition, subtraction and multiplication, but not under division: $1/2 \notin \mathbb{Z}$. The rationals are introduced precisely to allow quotients of integers with nonzero denominators. They are still not enough.

Theorem 1.1 (Irrationality of $\sqrt{2}$). $\sqrt{2}$ is irrational, i.e. $\sqrt{2} \notin \mathbb{Q}$.

The idea is to write $\sqrt{2}$ in lowest terms and show that 2 must nevertheless divide both numerator and denominator.

Proof. Suppose $\sqrt{2} \in \mathbb{Q}$, and write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring gives $m^2 = 2n^2$, so m^2 is even. An odd integer has an odd square, so m is even, say $m = 2\ell$ with $\ell \in \mathbb{Z}$. Then $2n^2 = 4\ell^2$, i.e. $n^2 = 2\ell^2$, and the same parity argument shows that n is even. So 2 divides both m and n , contradicting $\gcd(m, n) = 1$. $\square$

Assuming lowest terms turns the equation into a parity contradiction.

The real numbers that are not rational are the *irrational numbers* $\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}$.

Example. Every nonnegative real number has a *decimal representation*

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{N} \cup \{0\}, \quad x_i \in \{0, 1, \dots, 9\},$$

and negative real numbers are the negatives of such expansions. Some numbers have two decimal representations. For example, $1.999\dots = 2$: if $x = 1.999\dots$, then $10x = 19.999\dots$, so $9x = 18$ and $x = 2$.

Statements and quantifiers

Definition. A *statement* is a sentence with a definite truth value: it is either true or false.

The *universal quantifier* $\forall$ means “for all”, and the *existential quantifier* $\exists$ means “there exists”.

Example.

(i) “ $\forall x \in \mathbb{R}, x^2 - x \geq 0$ ” is false: it fails at $x = \frac{1}{2}$.

(ii) “ $\exists x \in \mathbb{R}, x^2 - x \geq 0$ ” is true: $x = 2$ works.

Notation. Quantifiers are read left to right. To negate a quantified statement, move the negation inwards and swap each quantifier:

$$\neg(\forall x \exists y P(x, y)) \iff \exists x \forall y \neg P(x, y).$$

Proposition. For every $x \in \mathbb{R}$ there exists $y \in \mathbb{R}$ such that $x^3 - y^3 = 1$.

Proof. Given x , take $y = \sqrt[3]{x^3 - 1}$. Then $x^3 - y^3 = x^3 - (x^3 - 1) = 1$. $\square$

Here y is allowed to depend on x ; this is exactly what the order “ $\forall x \exists y$ ” permits.

Proposition. The statement “ $\exists y \in \mathbb{R} \forall x \in \mathbb{R}, x^3 - y^3 = 1$ ” is false.

Proof. We prove its negation, $\forall y \in \mathbb{R} \exists x \in \mathbb{R}, x^3 - y^3 \neq 1$. Given y , take $x = y$; then $x^3 - y^3 = 0 \neq 1$. $\square$

Warning. Swapping $\forall$ and $\exists$ can turn a true statement into a false one. “ $\forall x \exists y$ ” asks for a possibly different y for each x ; “ $\exists y \forall x$ ” asks for a *single* y that works for every x at once. Above, the y that works is $\sqrt[3]{x^3 - 1}$, which genuinely depends on x .

1.2 Induction and well-ordering

Axiom (Principle of mathematical induction). Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If

- (i) $P(1)$ is true, and
- (ii) for every $k \in \mathbb{N}$, $P(k)$ implies $P(k + 1)$,

then $P(n)$ is true for every $n \in \mathbb{N}$.

To use it, verify the *base case* $P(1)$; then fix an arbitrary $k \in \mathbb{N}$, assume $P(k)$ (the *inductive hypothesis*) and deduce $P(k + 1)$.

Proposition. For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the displayed statement. $P(1)$ reads $1 = \frac{1 \cdot 2}{2}$. If $P(k)$ holds, then

$$\begin{aligned} 1 + 2 + \cdots + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}, \end{aligned}$$

which is $P(k+1)$. By induction, $P(n)$ holds for all n . $\square$

Induction has an equivalent formulation in terms of sets.

Proposition. Let $S \subseteq \mathbb{N}$. If $1 \in S$, and $k \in S$ implies $k+1 \in S$, then $S = \mathbb{N}$.

Proof. Apply induction to the statement $P(n)$: “ $n \in S$ ”. Then every $n \in \mathbb{N}$ lies in S , and $S \subseteq \mathbb{N}$, so $S = \mathbb{N}$. $\square$

The well-ordering principle

Theorem 1.2 (Well-ordering principle). Every nonempty subset S of $\mathbb{N}$ has a smallest element.

The idea is that if S had no smallest element, induction would show that S misses $1, 2, \dots, n$ for every n , so S would be empty.

Proof. Let $S \subseteq \mathbb{N}$ be nonempty, and suppose S has no smallest element. Let

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\},$$

the set of n such that S has “not yet been seen” by the time we reach n . We show $T = \mathbb{N}$ by induction.

Base case. If $1 \in S$, then 1 would be the smallest element of S , since it is the smallest element of $\mathbb{N}$. So $1 \notin S$, and $1 \in T$.

Inductive step. Suppose $k \in T$, i.e. $\{1, \dots, k\} \cap S = \emptyset$. If $k+1 \notin T$, then $\{1, \dots, k+1\} \cap S \neq \emptyset$, which by the inductive hypothesis forces $k+1 \in S$. But none of $1, \dots, k$ lies in S , so $k+1$ would be the smallest element of S , a contradiction. Hence $k+1 \in T$.

So $T = \mathbb{N}$. Then no natural number lies in S , i.e. $S = \emptyset$, contradicting that S is nonempty. $\square$

Well-ordering gives another way to prove statements about all n : look at the smallest counterexample.

Proposition. For every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof. Let $P(n)$ be the displayed statement, and suppose it fails for some n , i.e. $\neg(\forall n \in \mathbb{N}, P(n))$, equivalently $\exists n \in \mathbb{N}, \neg P(n)$. Then

$$S = \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty, so by the well-ordering principle it has a smallest element k . Since $P(1)$ holds, $k > 1$, so $k - 1 \in \mathbb{N}$, and $k - 1 \notin S$ by minimality. So $P(k - 1)$ holds:

$$1^2 + \cdots + (k - 1)^2 = \frac{(k - 1)k(2(k - 1) + 1)}{6} = \frac{(k - 1)k(2k - 1)}{6}.$$

Adding k^2 to both sides,

$$\begin{aligned} 1^2 + \cdots + k^2 &= \frac{(k - 1)k(2k - 1) + 6k^2}{6} = \frac{k((k - 1)(2k - 1) + 6k)}{6} \\ &= \frac{k(2k^2 + 3k + 1)}{6} = \frac{k(k + 1)(2k + 1)}{6}. \end{aligned}$$

So $P(k)$ holds, contradicting $k \in S$. Hence $S = \emptyset$. $\square$

Remark. This is the method of the *minimal counterexample*: if the set S of counterexamples is nonempty, take $k = \min S$; since k is the *first* failure, $P(k - 1)$ still holds, and one shows that $P(k - 1)$ forces $P(k)$. Induction and well-ordering are equivalent ways of expressing the same fact about $\mathbb{N}$: we proved well-ordering by induction, and here we used well-ordering to prove something that direct induction would also give.

Warning. The argument needs a *minimum*, not an infimum. Well-ordering applies to nonempty subsets of $\mathbb{N}$ and produces a least element that lies *in* S , which is what lets us conclude $\neg P(k)$. For a general subset of $\mathbb{R}$, an infimum need not belong to the set, so nothing could be concluded about $P(\inf S)$.

1.3 $\mathbb{R}$ as an ordered field

The real numbers form a *complete ordered field*. We build this up in three stages: the field axioms, which govern $+$ and $\cdot$; the order axioms, which let us compare elements; and the completeness axiom, which distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and is the source of all limiting behaviour in analysis.

The field axioms

Definition (Field). A *field* is a set F with two binary operations $+: F \times F \rightarrow F$ and $\cdot: F \times F \rightarrow F$ such that for all $a, b, c \in F$:

F1. (Associativity of addition) $a + (b + c) = (a + b) + c$.

F2. (Commutativity of addition) $a + b = b + a$.

F3. (Additive identity) There is $0 \in F$ with $a + 0 = a$.

F4. (Additive inverses) For every $a \in F$ there is $-a \in F$ with $a + (-a) = 0$.

F5. (Associativity of multiplication) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.

F6. (Commutativity of multiplication) $a \cdot b = b \cdot a$.

F7. (Multiplicative identity) There is $1 \in F$, $1 \neq 0$, with $a \cdot 1 = a$.

F8. (Multiplicative inverses) For every $a \in F$ with $a \neq 0$ there is $a^{-1} \in F$ with $a \cdot a^{-1} = 1$.

F9. (Distributive law) $a \cdot (b + c) = a \cdot b + a \cdot c$.

Example. $\mathbb{Q}$ also satisfies F1–F9. So being a field says nothing about size, order or completeness, and does not distinguish $\mathbb{R}$ from $\mathbb{Q}$.

Once the axioms are fixed, even the most familiar facts become theorems.

Theorem.

- (a) If $z, a \in \mathbb{R}$ and $z + a = a$, then $z = 0$.
- (b) If $u, b \in \mathbb{R}$, $b \neq 0$ and $u \cdot b = b$, then $u = 1$.
- (c) If $a \in \mathbb{R}$, then $a \cdot 0 = 0$.

Proof.

- (a) $z = z + 0 = z + (a - a) = (z + a) + (-a) = a + (-a) = 0$.
- (b) Multiplying $u \cdot b = b$ by b^{-1} , $u = u \cdot 1 = u \cdot (b \cdot b^{-1}) = (u \cdot b) \cdot b^{-1} = b \cdot b^{-1} = 1$.
- (c) $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$. Adding $-(a \cdot 0)$ to both sides and using associativity gives $0 = a \cdot 0$. $\square$

Remark. Every “obvious” manipulation is an axiom in disguise: moving a term across “=” uses additive inverses and associativity, cancelling a nonzero factor uses multiplicative inverses, and expanding a product uses the distributive law.

The order axioms

Definition (Order properties of $\mathbb{R}$). There is a nonempty subset $P \subseteq \mathbb{R}$, the set of *positive real numbers*, such that:

- O1.** (Trichotomy) For every $a \in \mathbb{R}$, exactly one of $a = 0$, $a \in P$, $-a \in P$ holds.
- O2.** (Closure under addition) If $a, b \in P$, then $a + b \in P$.
- O3.** (Closure under multiplication) If $a, b \in P$, then $a \cdot b \in P$.

Definition (Order relation). For $a, b \in \mathbb{R}$, we write $a > b$ (or $b < a$) if $a - b \in P$, and $a \geq b$ (or $b \leq a$) if $a - b \in P \cup \{0\}$.

So to show $a > b$, one shows $a - b \in P$; every manipulation of inequalities traces back to this and to O1–O3.

The positive rationals $P \cap \mathbb{Q}$ also satisfy O1–O3, so an ordered field still need not be $\mathbb{R}$. What $\mathbb{Q}$ lacks is completeness: the set $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$. This is why completeness is a separate axiom.

Axiom (Completeness axiom). If $A \subseteq \mathbb{R}$ is nonempty and bounded above, then A has a least upper bound in $\mathbb{R}$.

Upper bounds, suprema and the consequences of this axiom are developed in the next two subsections.

Example. If $a \in \mathbb{R}$ and $a \neq 0$, then $a^2 > 0$.

Proof. Since $a \neq 0$, trichotomy leaves two cases. If $a \in P$, then $a^2 = a \cdot a \in P$ by O3. Otherwise $-a \in P$, so $(-a)(-a) \in P$ by O3, and it remains to check $(-a)(-a) = a^2$. Multiplying $a + (-a) = 0$ by $-a$,

$$\begin{aligned} (-a)(a + (-a)) &= (-a) \cdot 0 = 0 \implies (-a)a + (-a)(-a) = 0 \\ &\implies (-a)(-a) = -((-a)a) = -(-a^2) = a^2. \end{aligned}$$

In either case $a^2 \in P$, i.e. $a^2 > 0$. $\square$

This is the typical use of trichotomy: split into the cases $a = 0$, $a \in P$, $-a \in P$, and use O2 and O3 to keep track of positivity.

Absolute value and the triangle inequality

Definition (Absolute value). For $a \in \mathbb{R}$, the *absolute value* of a is

$$|a| = \begin{cases} a & a \geq 0, \\ -a & a < 0. \end{cases}$$

Geometrically, $|a|$ is the distance from a to 0, and $|a - b|$ is the distance between a and b .

Theorem 1.3. For all $a, b \in \mathbb{R}$ and $c > 0$:

- (a) $|ab| = |a| |b|$;
- (b) $|a|^2 = a^2$;
- (c) $|a| \leq c \iff -c \leq a \leq c$;
- (d) $-|a| \leq a \leq |a|$.

Every part is proved the same way: split on the sign of a (and of b) and unwind the definition.

Proof. The definition gives $|-a| = |a|$ and $-|a| \leq a \leq |a|$, which is (d); checking the cases $a \geq 0$ and $a < 0$ gives (b).

For (a), if $a, b \geq 0$, then $ab \geq 0$ and $|ab| = ab = |a| |b|$. If $a \geq 0 > b$, then $ab \leq 0$ and $|ab| = -ab = a(-b) = |a| |b|$; the case $b \geq 0 > a$ is symmetric. If $a, b < 0$, then $ab > 0$ and $|ab| = ab = (-a)(-b) = |a| |b|$.

For (c), suppose $|a| \leq c$. If $a \geq 0$, then $a = |a| \leq c$ and $-c < 0 \leq a$. If $a < 0$, then $-a = |a| \leq c$, i.e. $-c \leq a$, and $a < 0 < c$. Conversely, suppose $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a \leq c$. If $a < 0$, multiplying $-c \leq a$ by -1 gives $|a| = -a \leq c$. $\square$

Proposition 1.4 (Triangle inequality). If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

The idea is to add the two-sided bounds of Theorem 1.3(d) for a and b , and then use part (c) to turn the result back into one inequality.

Proof. By Theorem 1.3(d), $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$. Adding,

$$-(|a| + |b|) \leq a + b \leq |a| + |b|.$$

If $|a| + |b| = 0$, this forces $a + b = 0$ and the claim is immediate. Otherwise $|a| + |b| > 0$, and Theorem 1.3(c) with $c = |a| + |b|$ gives $|a + b| \leq |a| + |b|$. $\square$

In terms of distance: going from 0 to $a + b$ directly is never longer than going via a .

Corollary. For all $a, b \in \mathbb{R}$:

$$(i) \quad ||a| - |b|| \leq |a - b|;$$

$$(ii) \quad |a - b| \leq |a| + |b|.$$

Proof.

- (i) Write $a = (a - b) + b$. The triangle inequality gives $|a| \leq |a - b| + |b|$, i.e. $|a| - |b| \leq |a - b|$. Symmetrically, $b = (b - a) + a$ gives $|b| - |a| \leq |b - a| = |a - b|$. Together,

$$-|a - b| \leq |a| - |b| \leq |a - b|,$$

which is the claim by Theorem 1.3(c).

$$(ii) \quad |a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|. \quad \square$$

Remark. The trick in (i) is to *add and subtract*: to bound $|a|$ in terms of $|a - b|$, write $a = (a - b) + b$ and apply the triangle inequality. It will be used constantly.

1.4 Suprema and infima

Bounds make sense in any ordered set. What is special about $\mathbb{R}$ is that every nonempty set that is bounded above has a supremum in $\mathbb{R}$; this need not hold in $\mathbb{Q}$.

Definition (Bounded sets). Let S be a nonempty subset of $\mathbb{R}$. Then S is

- (i) *bounded above* if there is $u \in \mathbb{R}$ (an *upper bound*) with $x \leq u$ for all $x \in S$;
- (ii) *bounded below* if there is $w \in \mathbb{R}$ (a *lower bound*) with $w \leq x$ for all $x \in S$;
- (iii) *bounded* if it is bounded above and below, and *unbounded* otherwise.

Definition (Supremum and infimum). An upper bound u_0 of S such that $u_0 \leq u$ for every upper bound u of S is the *supremum* (or *least upper bound*) of S , written $u_0 = \sup S$.

A lower bound w_0 of S such that $w_0 \geq w$ for every lower bound w of S is the *infimum* (or *greatest lower bound*) of S , written $w_0 = \inf S$.

The notation is justified because a set has at most one supremum: if u_1 and u_2 are both suprema of S , then $u_1 \leq u_2$, since u_1 is least and u_2 is an upper bound, and symmetrically $u_2 \leq u_1$. The same argument with the inequalities reversed shows that the infimum is unique.

The ϵ -characterization of the supremum

An upper bound u fails to be the *least* one exactly when some smaller number is still an upper bound, i.e. when there is a gap between S and u . So u is the supremum precisely when there is no gap: however small a margin $\epsilon > 0$ we allow, S has a point within ϵ of u .

Lemma 1.5 (ϵ -characterization of the supremum). An upper bound u of a nonempty set $S \subseteq \mathbb{R}$ is the supremum of S if and only if for every $\epsilon > 0$ there is $x_\epsilon \in S$ with $u - \epsilon < x_\epsilon$.

Proof. ($\Rightarrow$) Let $u = \sup S$ and $\epsilon > 0$. Since $u - \epsilon < u$ and u is the *least* upper bound, $u - \epsilon$ is not an upper bound. So some $x_\epsilon \in S$ satisfies $u - \epsilon < x_\epsilon \leq u$.

($\Leftarrow$) Suppose u is not least, so there is an upper bound $w < u$. Take $\epsilon = u - w > 0$. Then some $x_\epsilon \in S$ satisfies $w = u - \epsilon < x_\epsilon$, so w is not an upper bound after all. $\square$

So to show $u = \sup S$, one checks that u is an upper bound of S , and that for every $\epsilon > 0$ there is $x \in S$ with $u - \epsilon < x \leq u$. For $\inf S$, reverse the inequalities: a lower bound w such that every $\epsilon > 0$ has some $x \in S$ with $w \leq x < w + \epsilon$.

Example. Let $S = \{x \in \mathbb{R} : 0 \leq x < 1\}$. Then $\sup S = 1$ and $\inf S = 0$.

$\sup S = 1$: every $x \in S$ has $x < 1$, so 1 is an upper bound. Since S is nonempty and bounded above, $u = \sup S$ exists by completeness, and $u \leq 1$. Suppose $u < 1$, and let $\epsilon = \frac{1-u}{2} > 0$. Then

$$u < u + \epsilon = \frac{u+1}{2} < 1,$$

and $u + \epsilon \geq 0$ since $u \geq 0$ (as $0 \in S$). So $u + \epsilon \in S$ exceeds the upper bound u , a contradiction. Hence $u = 1$.

$\inf S = 0$: every $x \in S$ has $x \geq 0$, so 0 is a lower bound, and $w = \inf S \geq 0$. Every lower bound satisfies $w \leq 0$ because $0 \in S$. Hence $w = 0$.

Warning. $\sup S$ need not belong to S . Above, $\sup S = 1 \notin S$, so S has no maximum; $\inf S = 0 \in S$, so 0 is also the minimum. A supremum is a maximum only when it belongs to the set. Note also that “bounded above”, “bounded” and so on are properties of the *set* S , whereas $\sup S$ and $\inf S$ (when they exist) are *real numbers*.

1.5 Completeness and its consequences

We now use the completeness axiom seriously. Recall it once more.

Axiom (Completeness property). Every nonempty set of real numbers that has an upper bound has a supremum in $\mathbb{R}$.

Corollary. Every nonempty set of real numbers that has a lower bound has an infimum in $\mathbb{R}$.

Proof. Let $S \subseteq \mathbb{R}$ be nonempty with a lower bound w , so $w \leq s$ and hence $-w \geq -s$ for all $s \in S$. Then $S' = \{-s : s \in S\}$ is nonempty with upper bound $-w$, so by completeness $u = \sup S'$ exists. We claim $-u = \inf S$.

- Since $u \geq -s$ for all $s \in S$, we have $-u \leq s$ for all $s \in S$, so $-u$ is a lower bound for S .
- If w is any lower bound for S , then $-w$ is an upper bound for S' , so $u \leq -w$, i.e. $w \leq -u$.

So $-u$ is the greatest lower bound. $\square$

Example. Let $S = \{(1 + \frac{1}{n})^n : n \in \mathbb{N}\}$. By the binomial theorem,

$$\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1} \frac{1}{n} + \sum_{k=2}^n \binom{n}{k} \frac{1}{n^k} > 1.$$

For $0 \leq k \leq n$ we have $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} \leq \frac{n^k}{k!}$, and $k! \geq 2^{k-1}$ for $k \geq 2$, since each factor from 2 to k is at least 2. Hence

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = 3.$$

So $1 < (1 + \frac{1}{n})^n < 3$ for all n , and S is bounded.

The Archimedean property

We now show that $\mathbb{N}$ is not bounded above in $\mathbb{R}$. This uses two ingredients: completeness, which lets us form $u = \sup \mathbb{N}$ (as a subset of $\mathbb{R}$; nothing says $u \in \mathbb{N}$), and the fact that $\mathbb{N}$ is closed under $n \mapsto n + 1$, which produces a natural number beyond u .

Theorem 1.6 (Archimedean property). For every $x \in \mathbb{R}$ there is $n_x \in \mathbb{N}$ with $x \leq n_x$.

The idea is that if $\mathbb{N}$ had a supremum u , then $u - 1$ would not be an upper bound (Lemma 1.5 with $\epsilon = 1$), so some $m \in \mathbb{N}$ exceeds $u - 1$, and then $m + 1 \in \mathbb{N}$ lies beyond u .

Proof. Suppose not: there is $x \in \mathbb{R}$ with $x > n$ for all $n \in \mathbb{N}$. Then x is an upper bound for $\mathbb{N}$, and $\mathbb{N} \neq \emptyset$, so by completeness $u = \sup \mathbb{N}$ exists. Since $u - 1 < u$ and u is the *least* upper bound, $u - 1$ is not an upper bound for $\mathbb{N}$, so there is $m \in \mathbb{N}$ with $u - 1 < m$, i.e. $u < m + 1$. But $m + 1 \in \mathbb{N}$ and u is an upper bound for $\mathbb{N}$, so $m + 1 \leq u$. Contradiction. $\square$

Completeness + closure of $\mathbb{N}$ under $+1 \implies$ Archimedean property.

Corollary 1.7.

- (a) If $S = \{\frac{1}{n} : n \in \mathbb{N}\}$, then $\inf S = 0$.
- (b) If $t > 0$, there is $n_t \in \mathbb{N}$ with $0 < \frac{1}{n_t} < t$.
- (c) If $y > 0$, there is $n_y \in \mathbb{N}$ with $n_y - 1 \leq y < n_y$.

Each part is a single step: (a) follows from (b); (b) is the Archimedean property applied to $1/t$; and for (c) we take the *least* natural number to the right of y , which exists by well-ordering.

Proof.

- (a) 0 is a lower bound for S . If $b > 0$, then by (b) there is $n \in \mathbb{N}$ with $1/n < b$, and $1/n \in S$, so b is not a lower bound. Hence $\inf S = 0$.
- (b) By Theorem 1.6, choose $m \in \mathbb{N}$ with $1/t \leq m$. Then $n_t = m + 1 > 1/t$, so $0 < 1/n_t < t$.

- (c) Let $A = \{n \in \mathbb{N} : y < n\}$. By Theorem 1.6, $A \neq \emptyset$, so by the well-ordering principle (Theorem 1.2) it has a least element n_y , and $y < n_y$ since $n_y \in A$. Suppose $n_y - 1 > y$. Since $y > 0$, this forces $n_y \geq 2$, so $n_y - 1 \in \mathbb{N}$ and hence $n_y - 1 \in A$. But $n_y - 1 < n_y$, contradicting minimality. So $n_y - 1 \leq y < n_y$. $\square$

Warning. Part (c) is a *minimum* argument, not a supremum argument. The set $A = \{n \in \mathbb{N} : y < n\}$ is unbounded above, so $\sup A$ would not even be a real number. What we need is the least element of A , and well-ordering supplies it without any boundedness assumption. Contrast the supremum arguments for $\sup \mathbb{N}$ above and for $\sup S$ below, where the sets are subsets of $\mathbb{R}$ and boundedness above is essential.

The existence of $\sqrt{2}$

We know $\sqrt{2} \notin \mathbb{Q}$. Completeness should produce a square root of 2 in $\mathbb{R}$. Take the candidates from below, $S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}$; its supremum x ought to sit exactly on the boundary. There are only two ways it could miss. If $x^2 < 2$, then x can be nudged slightly to the right and stay in S , so x is not an upper bound. If $x^2 > 2$, then x can be nudged slightly to the left and remain an upper bound, so x is not the least one. Unlike Corollary 1.7(c), this is a genuine completeness argument.

Proposition 1.8. There is a real number x with $x^2 = 2$.

The nudge is by $1/n$. Working backwards, it suffices that $\frac{2x+1}{n} < 2 - x^2$ for the right nudge, and $\frac{2x}{n} < x^2 - 2$ for the left one; the Archimedean property provides such n .

Proof. Let $S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}$. Then $1 \in S$, and 2 is an upper bound, since $t > 2$ implies $t^2 > 4$. By completeness, $x = \sup S$ exists, and $x \geq 1$.

Case 1: $x^2 < 2$. Since $x \geq 1$ and $n \geq 1$,

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}.$$

As $2 - x^2 > 0$, the Archimedean property gives $n \in \mathbb{N}$ with $\frac{2x+1}{n} < 2 - x^2$. Then $\left(x + \frac{1}{n}\right)^2 < 2$, so $x + \frac{1}{n} \in S$, although $x + \frac{1}{n} > x = \sup S$. Contradiction. So $x^2 \geq 2$.

Case 2: $x^2 > 2$. Since $1/n^2 > 0$,

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n}.$$

As $x^2 - 2 > 0$, choose $n \in \mathbb{N}$ with $\frac{2x}{n} < x^2 - 2$. Then $\left(x - \frac{1}{n}\right)^2 > 2$. For any $t \in S$ we get $t^2 < 2 < \left(x - \frac{1}{n}\right)^2$, and $x - \frac{1}{n} \geq 0$ since $x \geq 1$; as $u \mapsto u^2$ is strictly increasing on $[0, \infty)$, this gives $t < x - \frac{1}{n}$. So $x - \frac{1}{n}$ is an upper bound for S smaller than $\sup S$. Contradiction. So $x^2 \leq 2$.

Hence $x^2 = 2$. $\square$

Density of $\mathbb{Q}$ and of the irrationals

Completeness is exactly what $\mathbb{Q}$ is missing. The rational version of the set above, $T = \{r \in \mathbb{Q} : r^2 < 2\}$, is nonempty ($0 \in T$) and bounded above in $\mathbb{Q}$ (every $r \in T$ has $r < 2$). In $\mathbb{R}$ it has a supremum $y = \sup T$, and the same two-case argument gives $y^2 = 2$, so $y = \sqrt{2}$. But $\sqrt{2} \notin \mathbb{Q}$, and y was the only possible least upper bound, so T has no supremum in $\mathbb{Q}$: $\mathbb{Q}$ is *not* complete. The last step needs the density of $\mathbb{Q}$ (Theorem 1.9 below). A rational least upper bound s of T would also be an upper bound in $\mathbb{R}$, so $s \geq \sqrt{2}$, and $s \neq \sqrt{2}$ since $\sqrt{2}$ is irrational. Density gives $r \in \mathbb{Q}$ with $\sqrt{2} < r < s$, and r is again an upper bound of T (every $t \in T$ has $t < \sqrt{2}$), so s was not least.

Still, $\mathbb{Q}$ is spread densely through $\mathbb{R}$. Lay down the multiples $\frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots$ on the number line; if the step $\frac{1}{n}$ is shorter than the interval (x, y) , the multiples cannot jump over it, so one of them lands inside.

Theorem 1.9 (Density of $\mathbb{Q}$). If $x, y \in \mathbb{R}$ and $x < y$, there is $r \in \mathbb{Q}$ with $x < r < y$.

For $0 < x < y$ we want $xn < m < yn$. We take m to be the first natural number past xn (Corollary 1.7(c)), so $m \leq xn + 1$, and this stays below yn as soon as $\frac{1}{n} < y - x$.

Proof. First let $0 < x < y$. By the Archimedean property there is $n \in \mathbb{N}$ with $\frac{1}{n} < y - x$, i.e. $xn + 1 < yn$. Since $xn > 0$, Corollary 1.7(c) gives $m \in \mathbb{N}$ with $m - 1 \leq xn < m$, i.e. $xn < m \leq xn + 1$. Hence

$$xn < m \leq xn + 1 < yn,$$

and dividing by $n > 0$ gives $x < \frac{m}{n} < y$ with $\frac{m}{n} \in \mathbb{Q}$.

If $x \leq 0$, choose $k \in \mathbb{N}$ with $-x < k$. Then $0 < x + k < y + k$, so by the first case there is $q \in \mathbb{Q}$ with $x + k < q < y + k$, and $r = q - k \in \mathbb{Q}$ satisfies $x < r < y$. $\square$

Corollary. If $x, y \in \mathbb{R}$ and $x < y$, there is an irrational z with $x < z < y$.

Proof. Let $a = x/\sqrt{2}$ and $b = y/\sqrt{2}$, so $a < b$. Choose a nonzero rational r with $a < r < b$: if $b \leq 0$ or $a \geq 0$, any rational in (a, b) given by Theorem 1.9 is nonzero; if $a < 0 < b$, take $r \in \mathbb{Q}$ with $0 < r < b$. Let $z = \sqrt{2}r$, so $x < z < y$. If z were rational, so would be $\sqrt{2} = z/r$, contradicting Theorem 1.1. $\square$

Corollary. If $x, y \in \mathbb{R}$ and $x < y$, there are infinitely many rationals between x and y .

Proof. Choose $r_1 \in \mathbb{Q}$ with $x < r_1 < y$. Given $r_k \in (x, y) \cap \mathbb{Q}$, apply Theorem 1.9 to $r_k < y$ to get $r_{k+1} \in \mathbb{Q}$ with $r_k < r_{k+1} < y$. The r_k are distinct rationals in (x, y) . $\square$

Countable and uncountable sets

Are there more rational or irrational numbers? To answer this we need to say in what sense one set is “bigger” than another.

Definition (Cardinality). A function $f : A \rightarrow B$ is a *bijection* if it is injective (one-to-one) and surjective (onto). The sets A and B have the same *cardinality*, written $|A| = |B|$, if there is a bijection $f : A \rightarrow B$.

Definition. A set S is *finite* if $S = \emptyset$ or there is a bijection $\{1, 2, \dots, n\} \rightarrow S$ for some $n \in \mathbb{N}$. It is *countably infinite* if there is a bijection $\mathbb{N} \rightarrow S$.

Example.

- (a) $f : \mathbb{N} \rightarrow \{2, 4, 6, \dots\}$, $f(n) = 2n$, is a bijection, so the even natural numbers have the same cardinality as $\mathbb{N}$.
- (b) $g : \mathbb{N} \rightarrow \mathbb{Z}$ given by $g(1) = 0$, $g(2k) = k$, $g(2k + 1) = -k$ ($k \in \mathbb{N}$) is a bijection, so $|\mathbb{Z}| = |\mathbb{N}|$.

Definition. A set is *countable* if it is finite or countably infinite, and *uncountable* otherwise.

2 Sequences and limits

We now define what it means for a sequence of real numbers to converge, and prove the first basic facts about limits: a limit is unique, and a convergent sequence is bounded.

2.1 Convergent sequences

Definition (Sequence). A *sequence* of real numbers is a function $X : \mathbb{N} \rightarrow \mathbb{R}$. We write $x_n = X(n)$ for its n th term and $\{x_n\}_{n=1}^{\infty}$, or simply $\{x_n\}$, for the sequence.

Definition (Convergence). A sequence $\{x_n\}$ *converges* to $a \in \mathbb{R}$ if for every $\epsilon > 0$ there is $n_\epsilon \in \mathbb{N}$ such that

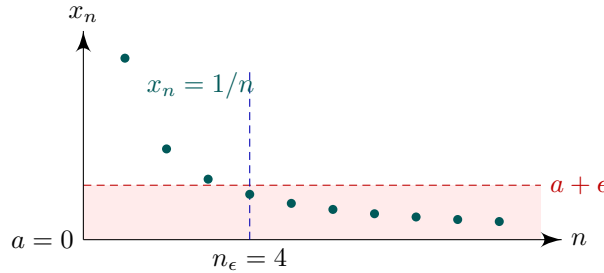
$$n \geq n_\epsilon \implies |x_n - a| < \epsilon.$$

We then write $\lim_{n \rightarrow \infty} x_n = a$, or $x_n \rightarrow a$, and call a the *limit* of the sequence.

In symbols,

$$x_n \rightarrow a \iff \forall \epsilon > 0 \quad \exists n_\epsilon \in \mathbb{N} \quad \forall n \geq n_\epsilon, \quad |x_n - a| < \epsilon.$$

Intuitively, for every tolerance $\epsilon > 0$, all sufficiently late terms lie in $(a - \epsilon, a + \epsilon)$. The index n_ϵ may depend on ϵ , and once $n \geq n_\epsilon$ the inequality must hold for *every* later term, even if earlier terms lie outside the interval. For $x_n = 1/n$ and $\epsilon = 0.3$, every term from $n_\epsilon = 4$ on lies in the shaded band around $a = 0$:



Definition (ϵ -neighbourhood). For $a \in \mathbb{R}$ and $\epsilon > 0$, the ϵ -neighbourhood of a is

$$V_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\} = (a - \epsilon, a + \epsilon).$$

So $x_n \rightarrow a$ precisely when for every $\epsilon > 0$ there is n_ϵ with $x_n \in V_\epsilon(a)$ for all $n \geq n_\epsilon$.

Example. $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. Given $\epsilon > 0$, the Archimedean property gives $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$, and then for every $n \geq n_\epsilon$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Example. $\lim_{n \rightarrow \infty} \frac{4n+3}{n+1} = 4$. The idea is to simplify the distance first: $\left| \frac{4n+3}{n+1} - 4 \right| = \frac{1}{n+1} < \frac{1}{n}$, so it is enough that $\frac{1}{n} < \epsilon$. Given $\epsilon > 0$, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$; then for every $n \geq n_\epsilon$,

$$\left| \frac{4n+3}{n+1} - 4 \right| = \frac{1}{n+1} \leq \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Uniqueness of the limit

Proposition 2.1. If $x, y \in \mathbb{R}$ and $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

Proof. If $x \neq y$, then $\epsilon = |x - y|/2 > 0$ but $|x - y| > \epsilon$. $\square$

Theorem 2.2. A sequence in $\mathbb{R}$ has at most one limit.

The idea is to pass from a to b through a late term x_n , making each of the two distances less than $\epsilon/2$.

Proof. Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$, and let $\epsilon > 0$. Choose $n_1, n_2 \in \mathbb{N}$ with $|x_n - a| < \epsilon/2$ for $n \geq n_1$ and $|x_n - b| < \epsilon/2$ for $n \geq n_2$. For $n \geq \max\{n_1, n_2\}$, the triangle inequality gives

$$|a - b| \leq |a - x_n| + |x_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

As $\epsilon > 0$ was arbitrary, $a = b$ by Proposition 2.1. $\square$

Divergence and boundedness

Definition. A sequence *diverges* if it does not converge to any real number.

Example. The sequence $0, 1, 0, 1, \dots$ ($x_n = 0$ for odd n , $x_n = 1$ for even n) diverges: both 0 and 1 keep appearing arbitrarily late, so the terms cannot all stay close to one limit. Indeed, suppose $x_n \rightarrow a$ and take $\epsilon = 1/2$. Choose an odd and an even index beyond n_ϵ . The odd term gives $|a| < 1/2$, i.e. $-1/2 < a < 1/2$, and the even term gives $|1 - a| < 1/2$, i.e. $1/2 < a < 3/2$. These intervals do not overlap.

Definition. A sequence $\{x_n\}$ is *bounded* if there is $M \in \mathbb{R}$ with $|x_n| \leq M$ for all $n \in \mathbb{N}$.

Theorem 2.3. Every convergent sequence of real numbers is bounded.

The idea is that the tail stays within 1 of the limit, and only finitely many terms come before it.

Proof. Suppose $x_n \rightarrow a$. With $\epsilon = 1$, choose n_1 such that $|x_n - a| < 1$ for all $n \geq n_1$. For these n , $|x_n| \leq |x_n - a| + |a| < 1 + |a|$. The remaining terms form a finite set, so

$$M = \max\{|x_1|, \dots, |x_{n_1-1}|, 1 + |a|\}$$

bounds every term (if $n_1 = 1$, take $M = 1 + |a|$). $\square$

The converse fails: $0, 1, 0, 1, \dots$ is bounded but diverges.

Example. The following sequences are unbounded: for every $M > 0$ some term has absolute value greater than M .

- (i) $a_n = \frac{n^2+1}{n}$: here $|a_n| = n + \frac{1}{n}$, and choosing $n_M \in \mathbb{N}$ with $n_M > M$ gives $|a_{n_M}| > M$.
- (ii) $a_n = (-1)^n n$: here $|a_n| = |(-1)^n| |n| = n$, which exceeds M as soon as $n > M$.

By Theorem 2.3, neither sequence converges.

Divergence to $\pm\infty$

Definition. A sequence $\{x_n\}$ *diverges to $+\infty$* , written $\lim_{n \rightarrow \infty} x_n = +\infty$, if for every $M > 0$ there is $n_M \in \mathbb{N}$ such that $x_n > M$ for all $n \geq n_M$. It *diverges to $-\infty$* , written $\lim_{n \rightarrow \infty} x_n = -\infty$, if for every $M > 0$ there is $n_M \in \mathbb{N}$ such that $x_n < -M$ for all $n \geq n_M$.

Warning. Although we write $\lim_{n \rightarrow \infty} x_n = \pm\infty$, such a sequence is still divergent: it does not converge to any real number.

Example.

$$(i) \lim_{n \rightarrow \infty} \frac{n^2}{1-n} = -\infty.$$

$$(ii) \lim_{n \rightarrow \infty} (\sqrt{n} + 1) = +\infty.$$

The proofs were not completed in lecture. For (ii), the suggested approach was to choose n so large that $\sqrt{n} > M - 1$.

Theorem. Suppose $\lim_{n \rightarrow \infty} x_n = a$ and $\lim_{n \rightarrow \infty} y_n = b$. Then for every $c \in \mathbb{R}$:

(i) the constant sequence $c, c, c, \dots$ converges to c ;

(ii) $\lim_{n \rightarrow \infty} cx_n = ca$.

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